Math 242, S09 Name: Answer

Quiz 7 SSN: xxx-xx-

Instructions: There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (10 points) Given the following homogeneous second-order differential equation:

$$4y'' + 8y' + 3y = 0$$

(a) Find the general solution of the given differential equation.

Solution: Characteristic equation.

$$4r^2 + 8r + 3 = 0$$

$$(2r+1)(2r+3) = 0$$

$$\Rightarrow r_1 = -\frac{1}{2}, \quad r_2 = -\frac{3}{2}$$

Thus, $y_1(x) = e^{-\frac{1}{2}x}$, $y_2(x) = e^{-\frac{3}{2}x}$

The general solution is

$$y(x) = C_1 y_1(x) + C_2 y_2(x)$$

$$= C_1 e^{-\frac{1}{2}x} + C_2 e^{-\frac{3}{2}x}$$

(b) Find a particular solution of the equation that satisfies the initial conditions

$$y(0) = 2, \quad y'(0) = -5.$$

Solution: $y(x) = C_1 e^{-\frac{1}{2}x} + C_2 e^{-\frac{3}{2}x}$

$$y'(x) = -\frac{1}{2}C_1 e^{-\frac{1}{2}x} - \frac{3}{2}C_2 e^{-\frac{3}{2}x}$$

$$\Rightarrow y(0) = C_1 + C_2 = 2$$

$$y'(0) = -\frac{1}{2}C_1 - \frac{3}{2}C_2 = -5$$

$\Rightarrow C_1 = -2$, $C_2 = 4$

Thus, $y(x) = -2e^{-\frac{1}{2}x} + 4e^{-\frac{3}{2}x}$